Hypothesis Testing

1. a. $X \sim \mathrm{B}(18, 0.25)$ $\mathrm{P}(X \leq 7) = 0.9431$

   b. $$X \sim \mathrm{B}(18, 0.25).p = 0.25.$$

   $$H_0 : p = 0.25$$
   $$H_1 : p > 0.25$$

   Test:

   $$P(X \geq 8) = 1 - P(X \leq 7) = 0.0570 > 0.05$$

   So there is not enough evidence to accept H_1 that the manufacturer has increased the proportion of *Saturns*.

2. $X \sim \mathrm{B}(20, 0.4)$. $p = 0.4$.

   $$H_0 : p = 0.4$$
   $$H_1 : p \neq 0.4$$

   Test:

   $$P(X \geq 12) = 1 - P(X \leq 11) = 0.0565 > 0.025$$
   $$P(X \leq 12) = 0.9790 > 0.025$$

   So there is not enough evidence to accept H_1 that less than or more than 40% of year 9 students signed up to the platform.

   Critical region:

   $$P(X \leq 4) = 0.0509 > 0.025$$
   $$P(X \leq 3) = 0.0159 < 0.025$$

   $$P(X \geq 12) = 1 - P(X \leq 11) = 0.0565 > 0.025$$
   $$P(X \geq 13) = 1 - P(X \leq 12) = 0.0210 < 0.025$$

   So critical region is $[0, 3] \cup [13, 20]$

3. $X \sim \mathrm{B}(12, 0.2)$

   a. $$H_0 : p = 0.2$$
   $$H_1 : p \neq 0.2$$

   b. **Test:**

   $$P(X \leq 0) = 0.0687 > 0.025$$
   $$P(X \geq 0) = 1 > 0.025$$

   So there is evidence to accept H_0 that one in every five tickets is a winner.

   c. 17 tickets.

   Then, $P(X \leq 0) = 0.0225 < 0.025$, and the critical region's lower tail is non-empty.

4. $X \sim \mathrm{B}(36, 0.2)$

a.
$$H_0 : p = 0.2$$
$$H_1 : p < 0.2$$

If $p < 0.2$, it means that the probability that bike tires have an unsafe tread has decreased, so their campaign has been successful because less tires have unsafe treads.

b. **Test:**

$$P(X \leq 1) = 0.0032 < 0.05$$

So there is not enough evidence to accept H_1 that the campaign has been successful.

c. The critical value is $X = 2$ because

$$P(X \leq 3) = 0.0522 > 0.05$$
$$P(X \leq 2) = 0.0160 < 0.05$$

So the critical region is $[0, 2]$ and 2 is the critical value.

d. A level of significance < 0.0032 would result in the opposite being concluded, because it would then mean that $P(X \leq 1)$ was greater than the significance level. This would mean that there is evidence to reject H_1 that the campaign has been successful.